Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/02/2013
PI Name:	Bhattacharya, Sharmila		
Project Title:	Gravity and oxidative stress responses in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	Sharmila.Bhattacharya@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Ames Research Center		
PI Address 1:	Building: N236, Room: 213		
PI Address 2:	MS 236-5		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	03/01/2010	**End Date:**	02/28/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 2/28/2014 per A. Chu/ARC (Ed., 3/4/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marcu, Oana (SETI Institute)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space environment of microgravity and radiation imposes stress conditions that affect the cellular and molecular responses of organisms and dictate their survival. A very fast response to stress is the production of reactive oxygen species (ROS), followed by intracellular oxidation. Space flight is associated with increased oxidative damage after return to 1 g, suggesting that microgravity and/or the transition between gravity fields triggers a response. We propose to use a model organism used previously for space flight, the fruit fly Drosophila melanogaster. Studies in both hyper- and microgravity have already established many of the physiological alterations, and work from our own laboratory suggests that oxidative stress is implicated in the molecular response under both conditions. We will use hypergravity as a test environment for gravity alterations, and to dissect out molecular and cellular responses as well as long-term physiological outcomes. The goal is to provide a complete picture of the oxidative stress pathway in hypergravity, and to determine and select the key molecular functions and physiological read-outs to be tested in microgravity. The aims are to quantify gravity-dependent cellular ROS production and to determine the key points of entry into the oxidative pathway by using mutants that show a physiologically altered response to hypergravity. This approach will give a picture of the stress pathway, starting from fast cellular changes triggered by sudden changes in the gravity vector, to tissue-specific gene and enzyme activity alterations. Understanding the stress pathway in hypergravity opens the venue into looking at the opposite transition in the gravity load, from 1 g to microgravity during space flight, and will help predict the cellular and physiological response to other stresses imposed by the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Some of the imaging tools developed to measure the behavioral changes for this project will be useful for a variety of behavioral studies in different organisms on Earth. The general issue of oxidative stress is also of great relevance to aging and stress on Earth, and the studies here therefore provide yet another paradigm in which to study and understand these molecular responses.
Task Progress:	The specific objectives achieved during this period were: 1. Quantified the behavioral changes during acute hypergravity exposure by deleting specific gene functions in the brain by RNA interference methodology (either knock-down function in the whole brain or targeted to specific regions of the brain), and also using stable ubiquitous loss-of-function mutants in similar gene pathways to study the same hypergravity induced changes. 2. Mapped out the molecular cues during acute hypergravity condition using gene expression studies by quantitative RT-PCR. 3. Built an optimal measurement system for monitoring real time behavior of flies in the centrifuge.
Bibliography Type:	Description: (Last Updated: 03/02/2021)
Abstracts for Journals and Proceedings	Hosamani R, Marcu O, Bhattacharya S. "Hypergravity-induced altered behaviour in Drosophila: Role of Parkinson's disease related genes." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 82. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Abstracts for Journals and Proceedings	Bhattacharya S, Chatterjee A, Marcu O, Globus RK. "Space Radiation and what we learn from small model organisms." International Conference of Radiation Biology, Mumbai, India, November 22-24, 2012. Invited talk. International Conference of Radiation Biology, Mumbai, India, November 22-24, 2012. , Nov-2012
Abstracts for Journals and Proceedings	Hagen B, Weisenhorn E, Pandey S, Angadi C, Bhattacharya S, Smith J. "Measuring the Behavioral Effects of Drosophila in Altered Gravity Environments." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 101. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Abstracts for Journals and Proceedings	Hosamani R, Marcu O, Wan J, Bhattacharya S. "Hypergravity-induced altered behaviour in Drosophila." Committee on Space Research (COSPAR) 2012 39th Scientific Assembly, Mysore, India, July 14-22, 2012. Committee on Space Research (COSPAR) 2012 39th Scientific Assembly, Mysore, India, July 14-22, 2012. Abstract F5.5-0009-12. , Jul-2012
Abstracts for Journals and Proceedings	Bhattacharya S. "Effects of Spaceflight in Drosophila melanogaster." 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. , Jun-2012
Articles in Peer-reviewed Journals	Chan KL, Inan O, Bhattacharya S, Marcu O. "Estimating the speed of Drosophila locomotion using an automated behavior detection and analysis system." Fly. 2012 Jul-Aug-Sep;6(3):205-10. http://dx.doi.org/10.4161/fly.20987 ; PMID: 22878427 , Jul-2012
Articles in Peer-reviewed Journals	Nigam Y, Knight J, Bhattacharya S, Bayer A. "Physiological changes associated with aging and immobility. Introduction to special issue." J Aging Res. 2012;2012:468469. http://dx.doi.org/10.1155/2012/468469 ; PubMed PMID: 22619717 , May-2012